

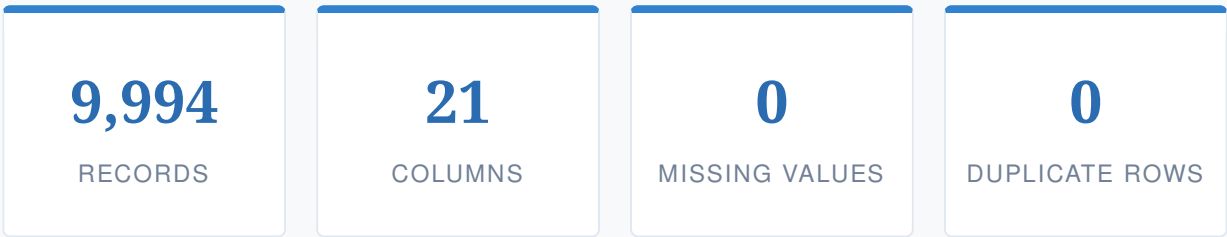


Exploratory Data Analysis

Superstore Dataset

Exploring sales, profitability, discounts, category performance, regional differences, and time trends.

Dataset Overview



Data Preparation

Prior to analysis, the raw dataset underwent essential preparation steps to ensure robust temporal and logistical evaluation:

- **Order Date** and **Ship Date** were converted to standard datetime formats.
- **Year** and **Month** attributes were derived directly from the Order Date to facilitate trend analysis.
- **Shipping Days** were computed by calculating the difference between the Ship Date and Order Date, providing a measure of fulfillment latency.

Sales & Profit Distribution

An evaluation of the foundational financial metrics reveals significant asymmetry in the data.

Sales Distribution

- Exhibits a **strong right-skew**.
- Concentrated in relatively low-value transactions.
- Contains prominent high-value Sales outliers.

Profit Distribution

- Exhibits a **strong right-skew** (skewness ≈ 7.56).
- Profit range spans from $\approx$ **-\$6,599 to \$8,399**.
- Contains both significant positive and negative Profit outliers.

Analytical Note: The presence of extreme high-value Sales and Profit outliers should not automatically be considered errors, as they may represent legitimate, albeit rare, high-volume or high-margin business transactions.

Key Relationships

Understanding the interplay between revenue generation and profitability is critical for strategic decision-making.

Sales vs. Profit

Correlation: 0.48

Demonstrates a **moderate positive relationship**. While higher Sales generally tend to be associated with higher Profit, there are notable instances where high-sales transactions still generate negative Profit.

Discount vs. Profit

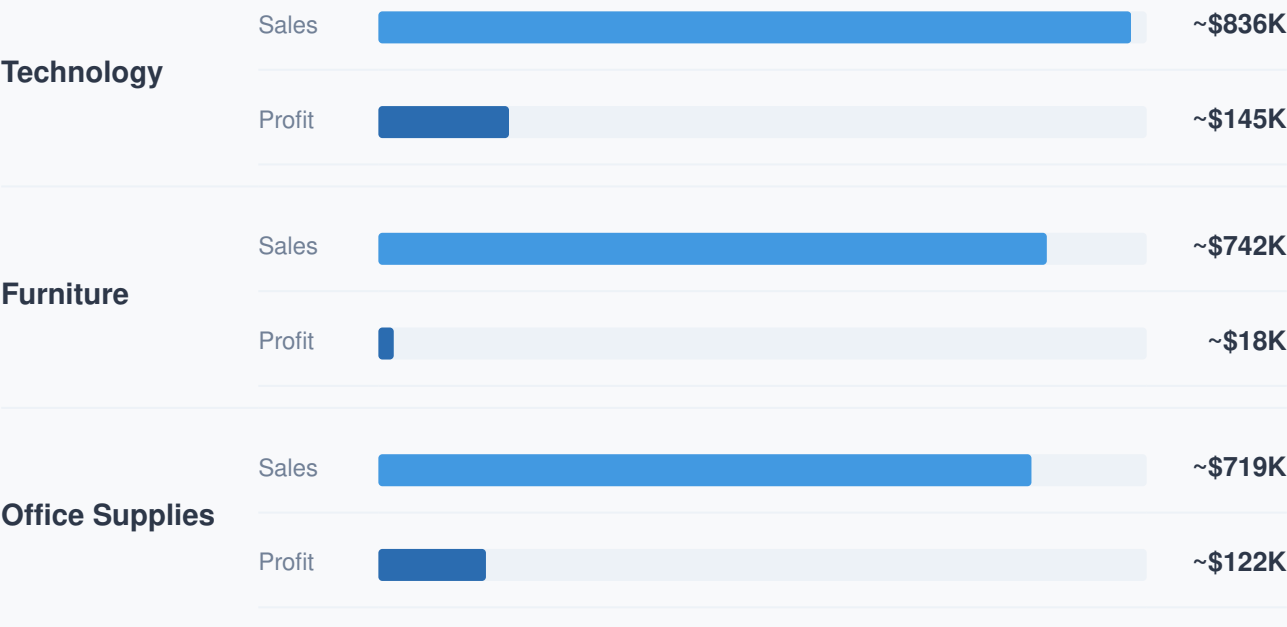
Correlation: -0.22

Demonstrates a **weak negative relationship**. The data suggests that higher discounts are generally associated with lower profitability across transactions.

Higher revenue does not automatically guarantee higher profitability.

Category Performance

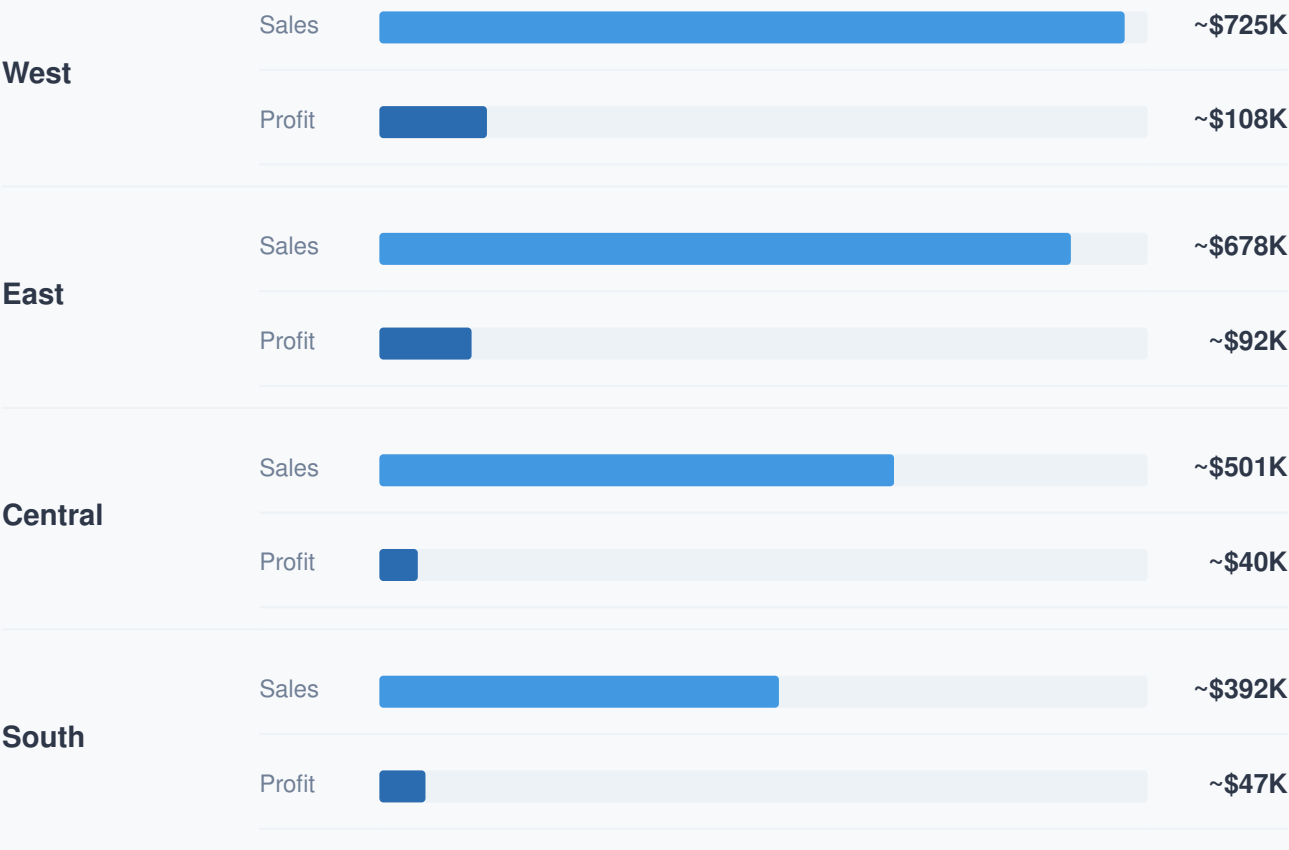
A comparative analysis of the primary product categories highlights distinct disparities between revenue generation and bottom-line profitability.



Profitability Concern: Furniture acts as a major profitability concern because it generates substantial Sales (~\$742K) but yields comparatively low Profit (~\$18K).

Regional Performance

Geographic analysis reveals uneven distribution of financial success across the four key regions.



West is the strongest overall region in both Sales and Profit. Conversely, Central generates more Sales than South but less Profit.

Time Trend Analysis

The dataset encompasses transactional records from **2014 to 2017**. Tracking metrics longitudinally yields the following observations:

- Sales fluctuate considerably over time, with multiple peaks and dips occurring throughout the four-year period.
- Profit exhibits greater volatility than Sales over the same time horizon.
- Distinct high-value sales spikes occur sporadically.
- Periods of negative-profit exist, indicating intervals where operational or discount strategies may have actively eroded margins.

Key takeaway: Sales growth does not always translate into stable profitability.

Correlation Analysis

To quantify the linear relationships between numeric variables, Pearson correlation coefficients were calculated. The matrix view underscores which factors most strongly associate with profitability.

Variable Pair	Correlation	Interpretation
Sales – Profit	0.48	Moderate positive relationship
Sales – Quantity	0.20	Weak positive relationship
Quantity – Profit	0.07	Very little linear relationship
Discount – Sales	-0.03	Negligible relationship
Discount – Profit	-0.22	Weak negative relationship

- **Sales** has the strongest relationship with Profit among the analyzed variables.
- **Discount** has a weak negative relationship with Profit, indicating a trade-off.
- **Quantity** has very little linear relationship with Profit, signifying volume does not dictate margins.

Key Findings Summary

Synthesizing the exploratory metrics yields ten fundamental insights:

1. Sales are strongly right-skewed and concentrated in relatively low-value transactions.
2. Profitability is highly variable and includes both positive and negative transactions.
3. Sales and Profit have a moderate positive relationship (0.48).
4. Discount and Profit have a weak negative relationship (-0.22).
5. Technology leads in both Sales and Profit.
6. Furniture generates substantial Sales but comparatively low Profit.
7. West is the strongest region in both Sales and Profit.
8. Central shows relatively weak profitability compared with its Sales volume.
9. Sales and Profit fluctuate considerably over time.
10. Sales volume alone does not determine profitability.

Business Insights & Conclusion

Translating statistical findings into operational perspective suggests several focal areas for strategic review.

Furniture Profitability

As the primary concern category, further investigation should focus on:

- Discount levels applied to these goods.
- Underlying product costs.
- Specific Furniture sub-categories dragging down margins.
- Products consistently yielding negative or low margins.

Discount Strategy

The negative Discount–Profit relationship suggests that excessive discounting may contribute to lower profitability. Adjustments to discount thresholds are advised.

Revenue vs Profit

Business performance should be evaluated using both Sales and Profit simultaneously, rather than relying on Sales volume alone as a proxy for success.

Regional Performance

While the West performs strongly, the Central region warrants further investigation because of comparatively weaker profitability against its realized sales volume.

Final Conclusion

The exploratory analysis reveals meaningful differences in sales distribution, profitability, discount behavior, category performance, regional performance, and time trends.

The strongest overall finding is that higher Sales generally support higher Profit, but Sales alone does not guarantee profitability.

Technology demonstrates strong overall performance, while Furniture represents the clearest profitability concern. West is the strongest region, whereas Central shows comparatively weaker profitability.

The weak negative relationship between Discount and Profit also indicates that discounting may contribute to reduced profitability.

Overall, the analysis demonstrates the importance of evaluating revenue, profitability, discounts, product categories, regions, and trends together rather than relying on Sales alone.